Original Article

Prevalence of Human Immunodeficiency Virus, Hepatitis B, Hepatitis C and Syphilis among Antenatal Patients, Gastroenterology Patients and Blood Donors in Public Hospitals in Lagos

Kehinde Sharafadeen Okunade[1], Emuobor Odeghe[2], Damilola O Akinlawon[3], Rita O Oladele[4]

Departments of [1]Obstetrics and Gynaecology, [2]Medicine, and [3]Community Health and Primary Care, Lagos University Teaching Hospital, [4]Department of Medical Microbiology and Parasitology, College of Medicine, University of Lagos, Lagos, Nigeria

Abstract

Background: Blood-borne infections remain a significant public health challenge globally. Understanding their prevalence and risk factors is crucial for implementing preventive measures. **Objective:** The objective of the study was to determine the prevalence of blood-borne infections, including human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and syphilis, among patients attending public hospitals in Lagos, Nigeria. **Subjects and Methods:** This was a cross-sectional study involving 352 participants, recruited from the antenatal, gastroenterology clinics and prospective donors from the blood banks in four general and one tertiary hospital in Lagos, Nigeria. Data were collected on demographic characteristics, risk factors and disease prevalence using structured questionnaires, and laboratory testing was performed using the B21147-01 HIV-HCV-HBsAg-Syphilis Combo Rapid Test. **Results:** The mean age of participants was 37.2 ± 11.0 years. The majority were female (66.8%) and married (80.7%). The prevalence of HIV, hepatitis B surface antigen (HBsAg) anti-HCV, and syphilis was 24.4%, 23.3%, 2.6% and 0.3%, respectively. There was significantly increased odds of HIV infection among participants with a history of sexually transmitted infections (adjusted odds ratio [AOR] = 2.3, 95% confidence interval [CI] = 1.0–5.1), and those who had skilled occupations (AOR = 4.5, 95% CI = 1.5–13.3). Participants with a family history of either HBV, HCV, chronic liver disease or hepatocellular carcinoma had significantly increased odds of being HBsAg positive (AOR = 7.8, 95% CI = 2.1–29.42). **Conclusion:** Blood-borne viral infections are highly prevalent among people attending hospitals in Lagos, Nigeria. Enhanced screening, awareness and risk reduction education are recommended to reduce the burden of these infections.

Keywords: Hepatitis B virus, hepatitis C virus, human immunodeficiency virus, prevalence, syphilis

Introduction

Infection with the human immunodeficiency virus (HIV) continues to be a significant global health challenge, with almost 40 million people living with the condition and about 42.3 million lives lost worldwide to the infection since the beginning of the epidemic.[1] Nigeria has about 190,950 HIV infections per year, the second-highest rate in the world. The country also has the world's second-highest burden of HIV/AIDS, with about 3 million people living with HIV (PLHIV).[2] In the 2018 Nigeria AIDS Indicator and Impact Survey, a large-scale, population-based survey, aimed at monitoring key national HIV indicators, adult HIV prevalence among individuals was estimated at 1.4%.[3] A more recent estimate after the pandemic showed a higher prevalence of 2.1%.[4]

The number of people living with chronic hepatitis B virus (HBV) infection worldwide was estimated by the

Received: 31-01-2025, **Revised:** 02-02-2025,
Accepted: 17-04-2025, **Published:** 14-05-2025

Access this article online

Quick Response Code:

Website: https://journals.lww.com/npmj

DOI: 10.4103/npmj.npmj_28_25

Address for correspondence: Dr. Damilola O Akinlawon, Department of Community Health and Primary Care, Lagos University Teaching Hospital, Lagos, Nigeria.
E-mail: damilolaakinlawon@gmail.com

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

How to cite this article: Okunade KS, Odeghe E, Akinlawon DO, Oladele RO. Prevalence of human immunodeficiency virus, hepatitis B, hepatitis C and syphilis among antenatal patients, gastroenterology patients and blood donors in public hospitals in Lagos. Niger Postgrad Med J 2025;32:127-34.

World Health Organisation (WHO) to be about 257 million in 2017. However, only about 10.5% of global infections have been diagnosed, and among these, only 17% are receiving treatment.[5] In addition, more than half a million deaths were recorded attributable to the disease during the same period.[5,6] The prevalence of chronic HBV infection in Nigeria is about 9.5%, translating to approximately 20 million individuals living with long-term HBV infection.[6] Nigeria is also among the countries with the highest burden of hepatitis C virus (HCV) infection worldwide, with an estimated prevalence of 1.4%.[7] However, routine diagnosis and treatment of the infection are not readily available in the country.[8]

Syphilis, caused by *Treponema pallidum* (TP), is a long-standing sexually transmitted infection (STI) linked to adverse pregnancy outcomes, increased HIV viral loads and reduced CD4+ T-cell counts in co-infected individuals. Its prevalence in Nigeria has been reported to range from 0.125% to 4.1% among pregnant women and commercial sex workers and from 0% to 2.63% among various populations with HIV in the country.[9] Although the prevalence of syphilis in Nigeria has been estimated to be lower than that in some other West African countries, the population involved is a cause of public health concern.[9] The Sustainable Development Goals (SDGs) aim to address the burden of HIV as a public health problem, primarily through SDG 3 (Good Health and Well-being), which targeted ending the AIDS epidemic by 2030.[10] In addition, it aims to tackle the social and economic factors that exacerbate vulnerability to diseases, promote universal health coverage, education, equitable access to preventive and treatment services and empowering women and girls, who are disproportionately affected by HIV.[11] A strategy that is needed to attain these goals is the adequate diagnosis of persons living with these infections, to reduce complications and disease spread. Shortly after the declaration of the SDGs, the WHO launched the Global Health Sector Strategy on Viral Hepatitis, aiming to eliminate viral hepatitis as a public health threat by 2030.[5] The strategy set ambitious targets, including diagnosing 90% of viral hepatitis infections by the year 2030, as part of its broader effort to reduce the global burden of infections.[5,12]

Accurate estimates of blood-borne infections are essential for tracking the dynamics of the epidemics, informing policy decisions, allocating resources effectively and guiding the development, implementation and evaluation of prevention and treatment programmes to improve population outcomes. Rapid point-of-care (POC) tests for these infections could significantly improve outcomes at the public health level by enabling early and simultaneous detection of multiple infections, ensuring timely treatment initiation and reducing disease progression, complications and transmission.[13]

These tests are particularly valuable in resource-limited settings as they offer accessible and cost-effective screening. Moreover, POC tests enhance linkage to care and reduce the burden of blood-borne infections, supporting the achievement of global health targets. The aim of this current study, therefore, was to determine the prevalence of HIV, HBsAg, anti-HCV and syphilis among patients attending public hospitals in Lagos, Nigeria utilising the B21147-01 HIV-HCV-HBsAg-Syphilis combo rapid POC test.

Materials and Methods

Study design and population

A descriptive cross-sectional study design was conducted to enrol patients from the antenatal, gastroenterology and prospective blood donor clinics of the blood banks of four general and one tertiary hospital in Lagos State. Ethical approval was received from the Health Research Ethics Committee of the Lagos University Teaching Hospital (approval number ADM/DSCST/HREC/APP/5428) prior to the commencement of the study. Permission was also obtained from the Lagos State Health Service Commission (HSC) before the engagement of participants. All participants provided written informed consent after the study was explained to them.

Sample size determination

Sample size determination was done using Cochran's formula; ($n = Z^2pq/d^2$), where: n is the minimum sample size, d = desired level of precision of the study at 5%, Z = standard normal deviate corresponding to 1.96 at 95% confidence interval (CI) and p = prevalence of chronic HBV in Nigeria = 9.5% = 0.095.[4] Therefore, with a nonresponse rate of 20%, the minimum sample size calculated was 159. A total of 352 participants were enrolled for the study.

Sampling technique

A multistage sampling approach was used to recruit study participants. In the first stage, purposive sampling was employed to select health facilities based on their capacity to provide relevant services for the study. This resulted in the inclusion of one tertiary hospital (Lagos University Teaching Hospital) and four general hospitals with blood bank services (Mushin, Orile-Agege, Ikorodu and Somolu General Hospitals). In the second stage, stratified random sampling (non-proportional) was used to categorise participants into three distinct groups: women attending the antenatal clinic, patients attending the gastroenterology clinic and prospective blood donors at the blood bank clinics. Each of the three strata was calculated to have a minimum of 53 participants (159/3) recruited. In the final stage, consecutive sampling was applied within each stratum, where eligible individuals meeting the study criteria were recruited sequentially until the required sample size was achieved. A minimum of 14 participants, who were prospective blood donors, were targeted to be selected (by consecutive sampling) from each of the general hospitals selected.

Data collection tool and procedure

Using a structured interviewer-administered questionnaire, data on sociodemographic characteristics and risk factors of HIV, HBV, HCV and syphilis were collected from each participant following informed written consent upon

explanation of the nature, purpose and procedure of the study. A venous blood sample was then obtained from each participant using a vacutainer sample receptacle to test for the blood-borne infections of interest using the POC B21147-01 HIV-HCV-HBsAg-Syphilis Combo Rapid Test. This POC test is a qualitative, easy-to-use, membrane-based chromatographic immunoassay for the detection of antibodies and antigens (antibodies to HIV type 1 and type 2, hepatitis B surface antigen [HBsAg], anti-HCV antibody and syphilis antibodies [IgG and IgM] to TP) in whole blood, serum or plasma that can be read visually without the need for an analyser. It is a cost-effective method for the diagnosis of HIV, HBV, HCV and syphilis infections. It is thus suitable for resource-limited settings such as Nigeria.

Each membrane is pre-coated with the relevant antigen or antibody in the test line regions. During testing, the whole blood, serum or plasma specimen reacts with recombinant antigens or specific antibodies. The mixture then migrates laterally on each of the membranes chromatographically by capillary action and reacts with recombinant antigen or antibody complexes on the membrane in the test line region.

Procedure for using test kit

1. Allow test cassette, specimen and/or controls to equilibrate to room temperature (15–30°C) prior to testing
2. Bring the pouch to room temperature before opening it. Remove the test cassette from the sealed pouch and use it as soon as possible. The best results will be obtained if the assay is performed within 1 h
3. Place the test cassette on a clean and level surface. Hold the dropper vertically and transfer 2 drops of the specimen (approximately 50 µl) to the specimen area each, then add 1 drop of buffer (approximately 40 µl), respectively. Start the timer
4. Wait for the coloured line(s) to appear. The test result should be read at 10 min. Results read after 20 min are considered invalid.

If the specimen contains the relevant antigen or antibody being tested for, one coloured line will appear in the test line region. This indicates a positive result for each test. If the specimen does not contain the relevant antigen or antibody being tested for, no coloured line will appear in the test line region indicating a negative result. To serve as a procedural control, a coloured line will always appear in the control line regions indicating that proper volume of specimen has been added and membrane wicking has occurred. The kit should be stored between 2°C and 30°C. The kits should not be frozen or stored in direct sunlight. The test is valid until the expiration date printed on the kit.

Data analysis

Data were entered into Microsoft Excel and analysed using International Business Machines (IBM) Statistical Package for the Social Sciences (SPSS) statistical software version 20 (IBM Corp., Armonk, New York, United States). The analysed data from the questionnaire were presented as frequency tables. Quantitative variables were described using appropriate measures of central tendency and dispersion. Logistic regression was used to determine the association between categorical variables using odds ratio at 95% CI, and $P < 0.05$ was taken for statistical significance.

Results

A total of 352 participants were enrolled in this study. Their mean age was 37.2 ± 11.0 years with a range of 18–82 years. More than two-thirds of the participants (66.8%) were female. The majority of the participants 284 (80.7%) were married. The highest proportion of participants 128 (36.4%) had professional occupations, and these were closely followed by those with unskilled occupations 122 (34.7%). More than half of the participants (58.3%) had tertiary education as their highest level of education [Table 1].

The prevalence of HIV, HBsAg anti-HCV and syphilis among the participants was 24.4%, 23.3%, 2.6% and 0.3%, respectively. Reduced patterns were similarly seen amongst the participants attending the antenatal clinic (19.7%,

Table 1: Sociodemographic characteristics of participants

Variable	Frequency (*n*=352), *n* (%)
Age group (years)	
18–29	89 (25.3)
30–39	138 (39.2)
40–49	81 (23.0)
≥50	44 (12.5)
Range	18–82
Mean±SD	37.2±11.0
Sex	
Male	117 (33.2)
Female	235 (66.8)
Marital status	
Single	54 (15.3)
Married	284 (80.7)
Divorced/widowed/separated	14 (4.0)
Occupation	
Professional	128 (36.4)
Skilled	60 (17.0)
Unskilled	122 (34.7)
Unemployed/retired	42 (11.9)
Education	
None	10 (2.8)
Primary	16 (4.5)
Secondary	121 (34.4)
Tertiary	205 (58.3)
Facility	
Secondary	105 (29.8)
Tertiary	247 (70.2)

SD: Standard deviation

Table 2: Prevalence of human immunodeficiency virus, hepatitis B virus, hepatitis C virus and syphilis, among participants

Infection*	Total (n=352), frequency (%)	Antenatal clinic (n=147), frequency (%)	Gastroenterology clinic (n=100), frequency (%)	Blood donors (n=105), frequency (%)
HIV	86 (24.4)	29 (19.7)	40 (40.0)	17 (16.2)
HBV	82 (23.3)	6 (4.1)	69 (69.0)	7 (6.7)
HCV	9 (2.6)	2 (1.4)	6 (6.0)	1 (1.0)
Syphilis	1 (0.3)	0	1 (1.0)	0
HIV/HBV co-infection	29 (8.2)	2 (1.4)	26 (26.0)	1 (1.0)
HIV/HCV co-infection	4 (1.1)	0	4 (4.0)	0
HBV/HCV co-infection	3 (0.9)	0	3 (3.0)	0

*Multiple responses allowed. HIV: Human immunodeficiency virus, HBV: Hepatitis B virus, HCV: Hepatitis C virus

Table 3: Risk factors of human immunodeficiency virus, hepatitis B virus, hepatitis C virus and syphilis among participants

Variable	Frequency (n=352), n (%)
Risk factors*	
History of STI	45 (12.8)
Alcohol intake	89 (25.3)
Blood transfusion	38 (10.8)
Smoking	24 (6.8)
Surgical history (local or orthodox)	62 (17.6)
Family history of HBV, HCV, CLD, HCC	26 (7.4)
Scarification/tattoo	30 (8.5)
Herbal concoction utilisation	96 (27.3)
Multiple sexual partners	29 (8.2)
Intravenous drug use	7 (2.0)

*Multiple responses allowed. HBV: Hepatitis B virus, HCV: Hepatitis C virus, CLD: Chronic liver disease, HCC: Hepatocellular carcinoma, STI: Sexually transmitted infection

4.1%, 1.4% and 0%, respectively) and the voluntary blood donors at the blood banks (16.2%, 6.7%, 1.0% and 0%, respectively) and an increased prevalence was seen among the participants attending the gastroenterology clinic (40.0%, 69.0%, 6.0% and 1.0%, respectively). The co-infection prevalence rates for HIV and HBsAg, HIV and anti-HCV and HBV and anti-HCV were 8.2%, 1.1% and 0.9%, respectively [Table 2].

The risk factors for blood-borne infections highlighted in the participants were consumption of herbal concoctions (27.3%), alcohol intake (25.3%), surgery (local or orthodox) (17.6%), a history of STIs (12.8%), blood transfusion (10.8%), smoking (10.8%), scarification/tattoo (8.5%), multiple sexual partners (8.2%), family history of chronic viral hepatitis, chronic liver disease (CLD) or hepatocellular carcinoma (HCC) (7.4%) and intravenous drug use (2.0%) [Table 3].

Symptoms that were present among the participants living with HIV were abdominal pain 12/86 (14.0%), weight loss 10/86 (11.6%), leg swelling 8/86 (9.3%), jaundice 6/86 (7.0%), altered sleep pattern 4/86 (4.7%), abdominal swelling 3/86 (3.5%), haematochezia 2/86 (2.3%), persistent or recurrent fever 1/86 (1.2%), melaena 1/86 (1.2%) and confusion/disorientation 1/86 (1.2%). Pallor was the most frequent examination finding seen in 11/86 (12.8%) among the participants living with HIV, and this was statistically significant (P = 0.001). Other statistically significant signs seen among the participants living with HIV included jaundice (7/86 [8.1%], P = 0.012) and abdominal tenderness (7/86 [4.1%], P = 0.001). Haematochezia (5/82 [6.1%], P = 0.006), abdominal swelling (4/82 [4.9%], P = 0.018) and persistent or recurrent fever (4/82 [4.9%], P = 0.018) were significant symptoms seen among participants living with HBV infection. Other symptoms seen among the participants living with HBV included weight loss 13/82 (15.9%), abdominal pain 9/82 (11.0%), jaundice 8/82 (9.8%), altered sleep pattern 4/82 (4.9%), haematemesis 2/82 (2.4%), melaena 2/82 (2.4%) and confusion/disorientation 1/82 (1.2%) [Table 4].

Pallor was a significant finding among participants with HBV infection (2/82 [2.4%], P = 0.020). Other signs seen among the HBV participants were hepatomegaly 8 (9.8%), jaundice 5 (6.1%), leg swelling 5/82 (6.1%), abdominal tenderness 3/82 (3.7%), ascites 2/82 (2.4%) and splenomegaly 1/82 (1.2%). Significant symptoms seen among participants with HCV infection included weight loss (3/9 [33.3%], P = 0.045) and altered sleep pattern (2/9 [22.2%], P = 0.048). Hepatomegaly 2/9 (22.2%), leg swelling 1/9 (11.1%), pallor 1/9 (11.1%), jaundice 1/9 (11.1%) and abdominal tenderness 1/9 (11.1%) were examination findings seen among participants with HCV infection though these were not statistically significant [Table 4].

Bivariate analysis showed that the factors associated with HIV infection were occupation (P = 0.013), the facility attended (P = 0.019), history of STI (P = 0.004), blood transfusion (P = 0.022), surgical history (local or orthodox) (P = 0.026) and lifestyle number of sexual partners (P = 0.005). Factors associated with HBV infection were sex (P < 0.001), occupation (P = 0.028), the facility attended (P = 0.019), family history of HBV, HCV, CLD or HCC (P = 0.029), history of consumption of herbal concoction (P < 0.001) and multiple sexual

Table 4: Prevalence of symptoms and signs among participants with human immunodeficiency virus, hepatitis B virus and hepatitis C virus infection

Variable	HIV, frequency (%) (*n*=86)	*P*/Fisher's exact	HBV, frequency (%) (*n*=82)	*P*/Fisher's exact	HCV, frequency (%) (*n*=9)	*P*/Fisher's exact
Symptoms						
Abdominal pain	12 (14.0)	0.056	9 (11.0)	0.561	2 (22.2)	0.245**
Weight loss	10 (11.6)	0.194	13 (15.9)	0.144	3 (33.3)	0.045***
Leg swelling	8 (9.3)	0.200	5 (6.1)	0.239	1 (11.1)	0.518**
Jaundice	6 (7.0)	0.193	8 (9.8)	0.107	0	1.000**
Altered sleep pattern	4 (4.7)	0.503**	4 (4.9)	1.000**	2 (22.2)	0.048***
Haematochezia	2 (2.3)	0.619**	5 (6.1)	0.006***	0	1.000**
Persistent or recurrent fever	1 (1.2)	1.000**	4 (4.9)	0.018***	0	1.000**
Abdominal swelling	3 (3.5)	0.066**	4 (4.9)	0.018***	1 (11.1)	0.131**
Melaena	1 (1.2)	1.000**	2 (2.4)	0.557**	0	1.000**
Confusion/disorientation	1 (1.2)	1.000**	1 (1.2)	1.000**	0	1.000**
Haematemesis	0	1.000**	2 (2.4)	0.136**	0	1.000**
Examination						
Leg swelling	8 (9.3)	0.428	5 (6.1)	0.106	1 (11.1)	0.571**
Pallor	11 (12.8)	0.001*	2 (2.4)	0.020***	1 (11.1)	0.480**
Hepatomegaly	7 (8.1)	0.138	8 (9.8)	0.265	2 (22.2)	0.095**
Jaundice	7 (8.1)	0.012***	5 (6.1)	0.554	1 (11.1)	0.324**
Abdominal tenderness	7 (8.1)	0.001***	3 (3.7)	1.000*	1 (11.1)	0.246**
Fever	1 (1.2)	1.000**	0	0.299**	0	1.000**
Ascites	2 (2.3)	0.076**	2 (2.4)	0.136**	0	1.000**
Finger clubbing	1 (1.2)	0.278**	0	1.000**	0	1.000**
Splenomegaly	0	1.000**	1 (1.2)	0.371**	0	1.000**

*Statistically significant, **Fisher's exact *P* value.***Both significant and Fisher's exact *P* value. HBV: Hepatitis B virus, HCV: Hepatitis C virus, HIV: Human immunodeficiency virus

partners ($P = 0.029$). HCV infection was not associated with either sociodemographic characteristics or any risk factor for bloodborne infections [Table 5].

Logistic regression analysis on associated variables revealed a significantly increased odds of participants with a history of STIs being a person living with HIV (adjusted odds ratio [AOR] = 2.291, 95% CI = 1.032–5.087, $P = 0.042$), and participants who had skilled (vocational) occupations had significantly increased odds of living with HIV than participants who had unskilled occupations (AOR = 4.522, 95% CI = 1.541–13.272, $P = 0.006$). Participants with a family history of HBV, HCV, CLD or HCC had significantly increased odds of having HBV than those with no history (AOR = 7.825, 95% CI = 2.080–29.445, $P = 0.002$). Furthermore, participants who were attending tertiary facilities had significant odds of HBV than those attending secondary facilities (AOR = 30.043, 95% CI = 11.257–80.183, $P < 0.001$) [Table 6].

Discussion

Blood-borne infections such as HIV, HBV, HCV and syphilis represent significant threats to blood safety and pose serious public health challenges. HIV is of particular concern due to its chronic carrier state and the severe complications associated with untreated or advanced infection, including immunosuppression and increased susceptibility to opportunistic diseases. The findings of this cross-sectional study highlight the significant burden of HIV, HBV, HCV and syphilis among patients attending public hospitals in Lagos.

The overall seroprevalence of HIV, HBsAg, anti-HCV and syphilis in this study was 24.4%, 23.3%, 2.6% and 0.3%, respectively. These values are higher than the estimates reported by large surveys in Nigeria.[3,4,6,7,9] The higher prevalence rates observed in this study underscore the importance of hospital-based studies in complementing community surveys. While community studies provide population-level estimates, hospital studies highlight the burden in symptomatic and at-risk populations, offering valuable insights for targeted interventions, resource allocation and policy formulation.[14]

Hospitals often attract patients with symptoms or advanced stages of diseases, including opportunistic infections, which may skew prevalence rates compared to community-based surveys that assess general population-level estimates. The higher prevalence of these infections observed in this study compared to the lower national estimates may be one of such, where individuals presenting may have a higher likelihood of having a broader spectrum of illnesses which may be associated directly or indirectly with blood-borne infections. Furthermore, hospital settings have higher

Table 5: Factors associated with human immunodeficiency virus, hepatitis B virus, hepatitis C virus infection among participants

Variable	HIV, frequency (%) (*n*=86)	*P*/Fisher's exact	HBV, frequency (%) (*n*=82)	*P*/Fisher's exact	HCV, frequency (%) (*n*=9)	*P*/Fisher's exact
Age group (years)						
18–29	20 (22.5)	0.147	16 (18.0)	0.076	3 (3.4)	0.226**
30–39	27 (19.6)		27 (19.6)		1 (0.7)	
40–49	24 (29.6)		25 (30.9)		3 (3.7)	
≥50	15 (34.1)		14 (31.8)		2 (4.5)	
Sex						
Male	32 (27.4)	0.369	51 (43.6)	0.000*	5 (4.3)	0.140**
Female	54 (23.0)		31 (13.2)		4 (1.7)	
Marital status						
Single	16 (29.6)	0.618**	18 (33.3)	0.145**	2 (3.7)	0.751**
Married	67 (23.6)		62 (21.8)		7 (2.5)	
Divorced/widowed/separated	3 (21.4)		2 (14.3)		0	
Occupation						
Professional	31 (24.2)	0.013*	39 (30.5)	0.028*	5 (3.9)	0.261**
Skilled	24 (40.0)		16 (26.7)		1 (1.7)	
Unskilled	24 (19.7)		18 (14.8)		1 (0.8)	
Unemployed/retired	7 (16.7)		9 (21.4)		2 (4.8)	
Education						
None	1 (10.0)	0.749**	0	0.252**	0	1.000**
Primary	3 (18.8)		2 (12.5)		0	
Secondary	32 (26.4)		30 (24.8)		3 (2.5)	
Tertiary	50 (24.4)		50 (24.4)		6 (2.9)	
Facility						
Secondary	17 (16.2)	0.019*	7 (6.7)	0.000*	1 (1.0)	0.289**
Tertiary	69 (27.9)		75 (30.4)		8 (3.2)	
Risk factors						
History of STI	15 (33.3)	0.004*	1 (2.2)	0.474**	1 (2.2)	0.447**
Alcohol intake	20 (22.5)	0.619	25 (28.1)	0.216	3 (3.4)	0.697**
Blood transfusion	15 (39.5)	0.022*	11 (28.9)	0.383	1 (2.6)	1.000**
Smoking	7 (29.2)	0.576	8 (33.3)	0.228	0	1.000**
Surgical history (local or orthodox)	22 (35.5)	0.026*	17 (27.4)	0.397	2 (3.2)	0.662**
Family history of HBV, HCV, CLD, HCC	12 (41.4)	0.078	16 (55.2)	0.029*	1 (3.4)	1.000**
Scarification/tattoo	12 (40.0)	0.107	12 (40.0)	0.719	1 (3.3)	1.000**
Herbal concoction utilisation	24 (25.0)	0.400	48 (50.0)	0.000*	3 (3.1)	1.000**
Multiple sexual partners	12 (41.4)	0.078	16 (55.2)	0.029*	1 (3.4)	1.000**
Lifetime sexual partners ≥3	29 (14.6)	0.005*	10 (5.1)	1.000**	2 (1.0)	0.517**
Intravenous drug use	3 (42.9)	0.368**	2 (28.6)	0.667**	0	1.000**

*Statistically significant, **Fisher's exact *P* value. HBV: Hepatitis B virus, HCV: Hepatitis C virus, CLD: Chronic liver disease, HCC: Hepatocellular carcinoma, STI: Sexually transmitted infection

testing rates due to routine screening protocols for specific groups, such as antenatal women or blood donors.[14] This increased diagnostic effort may uncover cases that would otherwise have remained undetected in community-based settings, likely contributing to the higher prevalence rates observed here. The components of rapid tests utilised in this study have been shown to have high sensitivity and specificity for the detection of various infections aiding the actualisation of effective estimations in the study population.[15]

These conditions eliciting a higher prevalence are also demonstrated in other hospital-based studies in the southeastern part of Nigeria among antenatal women that revealed the prevalence of HIV, HBV and HCV as 12.4%, 3.4% and 2.6%, respectively,[14] which were also above the national estimates for these infections. The prevalence of HIV, HBV and HCV in the study was also similar to that seen in our current study among the antenatal population which were at 19.4%, 4.1% and 1.4%, respectively. Another tertiary hospital-based study in south-south Nigeria revealed a higher prevalence of HIV, HBV, HCV and syphilis among prospective blood donors to be 4.2%, 4.1%, 3.6% and 3.1%, respectively.[16]

Various communities and regions may also have higher estimates of these infections influenced by different norms,

Table 6: Logistic regression of associated variables with human immunodeficiency virus and hepatitis B virus status

Variable	HIV AOR	95% CI	*P*	HBV AOR	95% CI	*P*
Sex						
Male	-	-	-	1.772	0.772–4.069	0.177
Female	-	-	-		Reference (1.0)	
Occupation						
Professional	1.095	0.456–2.628	0.839	0.512	0.170–1.544	0.234
Skilled	4.522	1.541–13.272	0.006*	0.644	0.187–2.217	0.485
Unskilled		Reference (1.0)			Reference (1.0)	
Unemployed/retired	0.549	0.111–2.722	0.463	1.223	0.267–5.614	0.795
Facility						
Secondary		Reference (1.0)			Reference (1.0)	
Tertiary	2.233	0.901–5.533	0.083	30.043	11.257–80.183	0.000*
History of STI						
Yes	2.291	1.032–5.087	0.042*	-	-	-
No		Reference (1.0)				
Blood transfusion history						
Yes	2.074	0.792–5.430	0.137	-	-	-
No		Reference (1.0)				
Surgical history (local or orthodox)						
Yes	1.516	0.630–3.646	0.353	-	-	-
No		Reference (1.0)				
Family history of HBV, HCV, CLD, HCC						
Yes	-	-	-	7.825	2.080–29.445	0.002*
No					Reference (1.0)	
Herbal concoction utilisation						
Yes	-	-	-	1.852	0.809–4.242	0.145
No					Reference (1.0)	
Multiple sexual partners						
Yes	-	-	-	1.688	0.551–5.172	0.360
No					Reference (1.0)	
Lifetime sexual partners ≥3						
Yes	1.925	0.883–4.198	0.099	-	-	-
No		Reference (1.0)				

*Statistically significant. AOR: Adjusted odds ratio, CI: Confidence interval, HBV: Hepatitis B virus, HCV: Hepatitis C virus, CLD: Chronic liver disease, HCC: Hepatocellular carcinoma

values and sociocultural practices operational in those populations. The prevalence figures reported in national studies showed a range of differences across communities, with states like Benue and Rivers having larger HIV population proportions of 5.7% and 5.2%, respectively.[4] Co-infection of viral hepatitis in PLHIV has been reported to be more frequent in rural Northern Nigeria and a recent study in that population revealed co-infection rates of HBV and HCV among PLHIV as high as 6.0% and 14.6%, respectively, and also HBV and HCV co-infection rates of 8.2% and 1.1%, respectively.[17]

Factors that were significantly associated with HIV infection in this study included a previous history of STI and skilled (vocational) occupations, while significant factors associated with HBV infection included a family history of HBV, HCV, chronic liver disease or hepatocellular carcinoma. Other studies have also highlighted various sociodemographic, socioeconomic and behavioural factors which may be linked to the advent of these infections.[6,8,16,17]

Overall, this study underscores the utility of rapid POC tests like the B21147-01 HIV-HCV-HBsAg-Syphilis Combo Test in simultaneously identifying multiple blood-borne infections particularly in resource-limited settings contributing to their attainment of global health targets such as the UNAIDS 95-95-95 goals and the WHO 2030 viral hepatitis elimination strategy. While this study provides valuable insights, a few limitations were noted. The cross-sectional design precludes establishing causal relationships between identified factors and these infections. In addition, the relatively small sample size may have limited the detection of significant associations. Future studies should employ longitudinal designs and larger sample sizes to better understand the dynamics of the occurrence of these infections and their important determinants.

Conclusion

There was a high prevalence of blood-borne infections observed amongst participants in this study, highlighting the urgent

need for strengthened public health interventions. Integrating rapid diagnostic testing, targeted screening and comprehensive care programmes is important to reducing the burden of these infections and improving population health outcomes.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. World Health Organization. HIV and AIDS; 2024. Available from: https://www.who.int/news-room/fact-sheets/detail/hiv-aids. [Last accessed on 2024 Nov 12].
2. Awofala AA, Ogundele OE. HIV epidemiology in Nigeria. Saudi J Biol Sci 2018;25:697-703.
3. Adeyinka DA, Olakunde BO, Oladimeji O, Ezeanolue EE. HIV indicator and impact survey: Considerations for Nigeria. Lancet HIV 2019;6:e348-50.
4. Onovo AA, Adeyemi A, Onime D, Kalnoky M, Kagniniwa B, Dessie M, *et al.* Estimation of HIV prevalence and burden in Nigeria: A Bayesian predictive modelling study. EClinicalMedicine 2023;62:102098.
5. World Health Organization. Global Hepatitis Report. Geneva: World Health Organization; 2017. Available from: https://www.who.int/hepatitis/publications/global-hepatitis-report2017/en/. [Last accessed on 2024 Nov 22].
6. Ajuwon BI, Yujuico I, Roper K, Richardson A, Sheel M, Lidbury BA. Hepatitis B virus infection in Nigeria: A systematic review and meta-analysis of data published between 2010 and 2019. BMC Infect Dis 2021;21:1120.
7. Dore GJ, Hatzakis A, Negro F, Waked I. Estimating HCV disease burden-volume 4 (editorial). J Viral Hepat 2017;24 Suppl 2:4-7.
8. Isaac WE, Ajani A, Jalo I, Akali Y, Oyeniyi C. Hepatitis C in adults and children: A cross-sectional review from a tertiary hospital, Northeast Nigeria during the period 2008-2015. J Infect Dis Epidemiol 2019;5:1-6.
9. Alubi PC, Okonko BJ, Ogbuji CC, Mbah EI, Okonko IO. Serological prevalence of herpes simplex virus, syphilis, and *H. pylori* co-infections amongst HIV-infected individuals receiving care in a secondary healthcare facility in Port Harcourt, Nigeria. J Multidiscip Res Updates 2023;5:1-16.
10. United Nations; Department of Economic and Social Affairs. Sustainable Development Goals (SDGs). Resolution Adopted by the General Assembly on 25 September 2015. United Nations; 2015. Available from: https://sdgs.un.org/goals. [Last accessed on 2024 Nov 30].
11. Halkos G, Gkampoura EC. Where do we stand on the 17 sustainable development goals? An overview on progress. Econ Anal Policy 2021;70:94-122.
12. World Health Organization. Global Health Sector Strategies on, Respectively, HIV, Viral Hepatitis and Sexually Transmitted Infections for the Period 2022-2030. Geneva: World Health Organization; 2022. Available from: https://www.worldhepatitisalliance.org/wp-content/uploads/2023/03/Global-Health-Sector-Strategy-Toolkit.pdf. [Last accessed on 2024 Dec 08].
13. Riegler AN, Larsen N, Amerson-Brown MH. Point-of-care testing for sexually transmitted infections. Clin Lab Med 2023;43:189-207.
14. Ikeako L, Ezegwui H, Ajah L, Dim C, Okeke T. Seroprevalence of human immunodeficiency virus, hepatitis B, hepatitis C, syphilis, and co-infections among antenatal women in a tertiary institution in South East, Nigeria. Ann Med Health Sci Res 2014;4:954-8.
15. Robin L, Mboumba Bouassa RS, Nodjikouambaye ZA, Charmant L, Matta M, Simon S, *et al.* Analytical performances of simultaneous detection of HIV-1, HIV-2 and hepatitis C-specific antibodies and hepatitis B surface antigen (HBsAg) by multiplex immunochromatographic rapid test with serum samples: A cross-sectional study. J Virol Methods 2018;253:1-4.
16. Okoroiwu HU, Okafor IM, Asemota EA, Okpokam DC. Seroprevalence of transfusion-transmissible infections (HBV, HCV, syphilis and HIV) among prospective blood donors in a tertiary health care facility in Calabar, Nigeria; an eleven years evaluation. BMC Public Health 2018;18:645.
17. Adesegun OA, Olaniran OH, Bamidele E, Inyang JN, Adegbe M, Binuyo TO, *et al.* HIV-hepatitis co-infection in a rural community in Northern Nigeria. Pan Afr Med J 2020;36:352.